Department of Mathematics
The Mathematics Community (TMC)

Introduction to Databases & SQL

DBMS, PostgreSQL & Core SQL Operations

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Outline

What is a Database?

Definition

A **database** is a structured, organised collection of data stored electronically in a computer system, designed for efficient storage, retrieval, and management.

Why choose a database over a spreadsheet?

Scalability
Handles massive data volumes that would crash or slow a spreadsheet.

Concurrency
Thousands of users can read and write simultaneously without conflicts.

Data Integrity
Built-in rules keep data accurate and consistent at all times.

Security
Fine-grained access controls — only authorised users can view or edit data.

What is a DBMS?

Definition

A **Database Management System (DBMS)** is software that enables users to **create, define, maintain, and control** access to a database. It acts as the interface between the user and the stored data, ensuring data remains consistently organised and accessible.

Relational (SQL)

- Data stored in structured **tables** (rows & columns)
- Tables linked via **keys**
- › PostgreSQL, MySQL, Oracle

Non-Relational (NoSQL)

- Flexible formats: documents, key-value pairs, graphs
- Suited for unstructured or widely varied data
- › MongoDB, Redis, Firebase, Supabase

This seminar focuses on **Relational Databases** and the language used to interact with them — **SQL**.

SQL & CRUD

SQL — Structured Query Language

The **standard language** for communicating with relational databases. SQL lets you perform four fundamental operations, collectively called **CRUD**:

Letter	Operation	SQL Command	Purpose
C	Create	`INSERT`	Add new records
R	Read	`SELECT`	Retrieve records
U	Update	`UPDATE`	Modify existing records
D	Delete	`DELETE`	Remove records

NB: All SQL statements must end with a semicolon “ ; ”

Create a database:

```
CREATE DATABASE school;
```

Create a table to store student records:

```
CREATE TABLE students (
    id          SERIAL PRIMARY KEY,
    fullname    VARCHAR(50) NOT NULL,
    age         INTEGER,
    stage       INTEGER
);
```

Data Types

- `SERIAL` — auto-incrementing integer
- `VARCHAR(n)` — string up to *n* characters
- `INTEGER` — whole number

Constraints

- `PRIMARY KEY` — uniquely identifies each row
- `NOT NULL` — field is required; cannot be empty

Installing PostgreSQL — Windows

Installation Steps

1. Go to postgresql.org/download/windows
2. Download the **Windows x86-64** installer
3. Run the .exe, grant permissions, click *Next*
4. Set a password for the default superuser postgres
5. Keep the default port **5432** and finish the wizard

Verify & Connect

```
psql --version
```

```
psql -U postgres
```

Tip

Keep the postgres superuser password safe — you will need it every time you connect.

Install PostgreSQL and the contrib extension package:

```
sudo apt update && sudo apt full-upgrade -y
sudo apt install postgresql postgresql-contrib -y
```

Start and enable the service to run on boot:

```
sudo systemctl start postgresql
sudo systemctl enable postgresql
```

Check that the service is running:

```
sudo systemctl status postgresql
```

Switch to the default PostgreSQL user and connect:

```
sudo -i -u postgres
psql
```

You are now inside the `psql` interactive terminal.

Setting Up a mock Database

Create and connect to the database

```
CREATE DATABASE my_database;
\c my_database
```

Create the students table with a default value on `enrolled`

```
CREATE TABLE students (
    id       SERIAL PRIMARY KEY,
    name     VARCHAR(100) NOT NULL,
    email    VARCHAR(150) NOT NULL,
    age      INT,
    enrolled BOOLEAN DEFAULT TRUE
);
```

Inserting Records (DML — INSERT)

Insert records one at a time

```sql
INSERT INTO students (name, email, age, enrolled)
VALUES ('James Nkrumah', 'jamesnkrumah@gmail.com', 21, TRUE);

INSERT INTO students (name, email, age, enrolled)
VALUES ('Yaw Boadi', 'yawb@gmail.com', 24, FALSE);
```

Insert multiple records in a single query

```sql
INSERT INTO students (name, email, age)
VALUES
    ('Prince Adomakoe',   'prince123@gmail.com',    19),
    ('Lawrencia Nyarko', 'lawrencia201@gmail.com', 20),
    ('Alice Mensah',      'mensahalic@gmail.com',   21),
    ('Edith Agyapong',    'edith.agy@gmail.com',    22),
    ('Kofi Asante',       'asanteyoyo@gmail.com',   19);
```

Reading Data (DML — SELECT)

Reading records from a table

```sql
-- All columns and all rows
SELECT * FROM students;
-- PostgreSQL is case-insensitive: STUDENTS == students

-- Specific columns only
SELECT name, email FROM students;

-- Filter with a condition
SELECT name, email FROM students
WHERE enrolled = TRUE;

-- Sort results alphabetically
SELECT * FROM students ORDER BY name ASC;

-- Return only the first 5 rows
SELECT * FROM students LIMIT 5;
```

Updating & Deleting Records

UPDATE — change an existing value

```
UPDATE students
SET age = 22
WHERE name = 'Kofi Asante';
```

DELETE — remove a specific row

```
DELETE FROM students
WHERE name = 'Alice Mensah';
```

NB: Always include a `WHERE` **clause** with both `UPDATE` and `DELETE`. Without it, **every row** in the table will be modified or deleted.

`psql` Meta-Commands — Quick Reference

Command	Description
`\l`	List all databases
`\c dbname`	Connect to / switch database
`\dt`	List all tables in current database
`\d tablename`	Describe a table's structure
`\du`	List all users and roles
`\q`	Quit the `psql` terminal

A complete cheatsheet for **PostgreSQL** could be found here:
`obedadu-gyamfi.github.io/.../psql-cheatsheet.html`

Summary

Concepts Covered

- What a database is and why it beats a spreadsheet
- RDBMS (SQL) vs NoSQL — and when each fits
- SQL as the standard interface to relational data
- CRUD — the four core data operations
- DDL vs DML — structure vs data
- Installing PostgreSQL on Windows and Linux

Key Commands

- `CREATE TABLE` — define structure
- `ALTER TABLE` — modify structure
- `INSERT INTO` — add data
- `SELECT ... WHERE` — query data
- `UPDATE ... SET` — modify data
- `DELETE FROM` — remove data

Explore Next

`JOIN`s, foreign keys, indexes, transactions, user roles, and pgAdmin.

Access all the resources and more information here!.

- www.github.com/obedAdu-Gyamfi/TMC
- postgresql.org
- pgadmin.org/download
- psql-cheatsheet
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Thank You!